

Galaxy Scaling Relations from $z \approx 0$ to the JWST Frontier: Evolution of the Star-Forming Main Sequence and the Mass–Metallicity Relation

NEBULAMIND AUTONOMOUS RESEARCH PIPELINE¹

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ABSTRACT

We anchor two fundamental galaxy scaling relations — the star-forming main sequence (SFMS) and the stellar mass–gas-phase-metallicity relation (MZR) — at $z \approx 0$ using $\sim 5 \times 10^5$ SDSS galaxies, and confront them with JWST/NIRSpec spectroscopy of $3 < z < 9$ galaxies drawn from the Nakajima et al. (2023) census and the MIRI/CEERS SED catalogue of Lisiecki et al. (2025). Relative to the local sequence at fixed stellar mass, high-redshift galaxies are elevated in star-formation rate by +0.77 dex at $z \approx 3.5$, rising monotonically to +1.9 dex at $z \gtrsim 6$, consistent with a specific-SFR that increases steeply with cosmic time. In metallicity they lie ≈ 0.4 dex below the local MZR at fixed mass, measured against a T_e -anchored local abundance scale (PP04 O3N2) chosen to match the direct- T_e high- z metallicities. This metallicity deficit is nearly *constant* (−0.43 to −0.37 dex) across $z \approx 4$ –7, rather than deepening with redshift, and the population shows large scatter that includes individual systems approaching the local relation. Both signatures — a flat high- z metallicity offset and enriched outliers — bear directly on the current debate over whether early galaxies are as metal-poor as hierarchical chemical-evolution models predict. We describe caveats (heterogeneous emission-line selection, calibration systematics, extrapolation of the local relations to low mass) and outline a companion simulation-versus-observation test using IllustrisTNG. This manuscript was generated autonomously; all numbers are reproducible from the public queries listed.

Keywords: Galaxy evolution — Galaxy chemical evolution — Star formation — High-redshift galaxies

1. INTRODUCTION

The star-forming main sequence (a tight, roughly power-law relation between stellar mass $M_\star$ and star-formation rate, SFR; Noeske et al. 2007; Speagle et al. 2014) and the mass–metallicity relation (Tremonti et al. 2004; Kewley & Ellison 2008) are among the most economical summaries of how galaxies build their stars and metals. Their *evolution* encodes the changing balance of gas accretion, star formation, and outflows over cosmic time. JWST has for the first time extended these relations with rest-frame optical spectroscopy into the reionization epoch, and the early results are contested: some analyses find galaxies markedly metal-poor at fixed mass as expected, while others report surprisingly rapid early enrichment (Nakajima et al. 2023; Curti et al. 2024; Sanders et al. 2021). Our automated research pipeline, which draws candidate frontiers bottom-up from a large literature corpus, flagged high-redshift nebular diagnostics as the fastest-growing and most contested frontier. Here we act on that by measuring, with a uniform lo-

cal anchor, how far JWST galaxies have moved off the $z \approx 0$ relations.

2. DATA

2.1. The $z \approx 0$ anchor: SDSS

We use the MPA–JHU value-added catalogue derived from SDSS spectroscopy (Brinchmann et al. 2004; Tremonti et al. 2004), queried live from the SkyServer (galSpecExtra: total stellar mass `lgm_tot_p50`, SFR `sfr_tot_p50`, specific SFR `specsfr_tot_p50`, and, for the MZR, the star-forming subsample with $12 + \log(\text{O}/\text{H})$). After quality cuts we retain $N = 4.9 \times 10^5$ galaxies over $8.75 < \log M_\star < 11.75$. Fitting the star-forming population ($\log \text{sSFR} > -11$) gives a main sequence

$$\log \text{SFR} = 0.61 (\log M_\star - 10) + 0.065, \quad \sigma \approx 0.39 \text{ dex}, \quad (1)$$

and an asymptotic MZR (Moustakas et al. 2011) of the form $12 + \log(\text{O}/\text{H}) = Z_0 - \log[1 + (10^{M_0 - \log M_\star})^\gamma]$ with

$(Z_0, M_0, \gamma) = (9.22, 9.997, 0.524)$. These define our fixed reference relations.

2.2. The high- z frontier: JWST

For $3 < z < 9$ we combine two public JWST catalogues obtained from the VizieR TAP service. (i) [Nakajima et al. \(2023\)](#) (VizieR J/ApJS/269/33): 180 galaxies with NIRSpec spectroscopic redshifts, SED stellar masses, SFRs, and direct/strong-line $12+\log(\text{O}/\text{H})$ (145 with metallicity), spanning $z = 3.8\text{--}8.9$. (ii) [Lisiecki et al. \(2025\)](#) (VizieR J/A+A/708/A235): 3743 galaxies at $z = 3\text{--}6$ with MIRI/CEERS SED masses and SFRs (no metallicity), used to boost main-sequence statistics. Masses are placed on a Chabrier/Kroupa scale, consistent with the SDSS anchor to within ~ 0.03 dex.

3. METHOD

For every high- z galaxy we compute its offset from the *local* relation at its own stellar mass: $\Delta \log \text{SFR} \equiv \log \text{SFR} - \text{SFMS}_{z \approx 0}(\log M_\star)$ and $\Delta_{\text{O}/\text{H}} \equiv [12 + \log(\text{O}/\text{H})] - \text{MZR}_{z \approx 0}(\log M_\star)$. We report the median and 16–84th percentile of these offsets in redshift bins. This differential approach cancels the (identical) mass axis and isolates evolution in the SFR and metallicity directions. Critically, to avoid an abundance-scale mismatch, the MZR anchor is recomputed on a T_e -anchored strong-line scale (PP04 O3N2) from SDSS `galSpecLine` fluxes ($N = 2.0 \times 10^5$), matched to the direct- T_e scale on which the high- z metallicities are derived; the earlier photoionization-model (Tremonti) anchor lies ~ 0.24 dex higher and by itself overstated the deficit by $\sim 0.1\text{--}0.13$ dex.

4. RESULTS

Main sequence. High- z galaxies are elevated above the local sequence at fixed mass by $+0.77$ dex ($z \approx 3.5$), $+0.89$ dex ($z \approx 4.7$), $+0.96$ dex ($z \approx 5.4$), and $+1.94$ dex ($z \approx 6.7$; Fig. 1 left, Fig. 2 left). A factor of ~ 6 enhancement in SFR at fixed mass by $z \gtrsim 6$, is the expected consequence of higher gas fractions and shorter gas-consumption timescales in the early Universe. The highest-redshift bin ($z > 6$, $n = 46$) should be read cautiously given the small sample and the strong emission-line selection.

Mass–metallicity relation. At fixed mass, high- z galaxies are metal-poor relative to $z \approx 0$ by -0.43 dex ($z \approx 4.6$), -0.37 dex ($z \approx 5.3$), and -0.40 dex ($z \approx 7.2$; Fig. 2 right), all on the T_e -anchored scale. Two features stand out. First, the deficit is *nearly constant* from $z \approx 4$ to $z \approx 7$ rather than deepening — early galaxies

are not progressively more pristine with lookback time in this sample. Second, the scatter is large (16–84th range spanning ~ 0.5 dex) and includes systems within ~ 0.15 dex of the local relation. Both are directly relevant to the ongoing debate over “surprisingly enriched” early galaxies: a flat offset plus enriched outliers is more consistent with rapid early enrichment followed by a slowly evolving equilibrium than with monotonic metal build-up.

5. DISCUSSION AND CAVEATS

The measurements above are differential and therefore robust to the absolute calibration of the local relations, but several systematics temper interpretation. (1) The JWST samples are emission-line selected and thus biased toward high-sSFR, high-EW systems, which inflates the apparent main-sequence elevation and may bias metallicities. (2) We mitigate the dominant abundance-scale systematic by placing the local anchor on the same T_e -anchored scale as the high- z data; a residual ~ 0.1 dex uncertainty between direct- T_e and T_e -anchored strong-line calibrations remains, and high-ionization conditions at early times may add scatter. (3) We extrapolate the SDSS relations below $\log M_\star \approx 8$ to overlap the low-mass JWST galaxies; the main-sequence extrapolation is mild (linear) but the MZR extrapolation is more uncertain. (4) Aperture and IMF differences are sub-dominant ($\lesssim 0.05$ dex). (5) The $z > 6$ bins are small. A natural next step — enabled by the same pipeline — is to confront these observed relations with a cosmological simulation (IllustrisTNG) processed identically, testing whether a model tuned to $z \approx 0$ reproduces the *evolution* we measure. That “calibration is not validation” test is the single highest-ranked frontier in our topic map and is deferred to a companion study.

6. CONCLUSION

Using a uniform SDSS anchor and public JWST catalogues, we find that $3 < z < 9$ galaxies are elevated 0.8–1.9 dex above the local star-forming main sequence and depressed ≈ 0.4 dex below the local (T_e -anchored) mass–metallicity relation, with the metallicity deficit remaining flat across $z \approx 4\text{--}7$. The combination of a non-deepening metallicity offset with enriched outliers favours rapid early enrichment toward an evolving equilibrium, and provides a concrete, reproducible target for cosmological simulations. This study was produced autonomously as a demonstration of literature-to-data frontier research; the SDSS SkyServer and VizieR TAP queries are public and the analysis is fully reproducible.

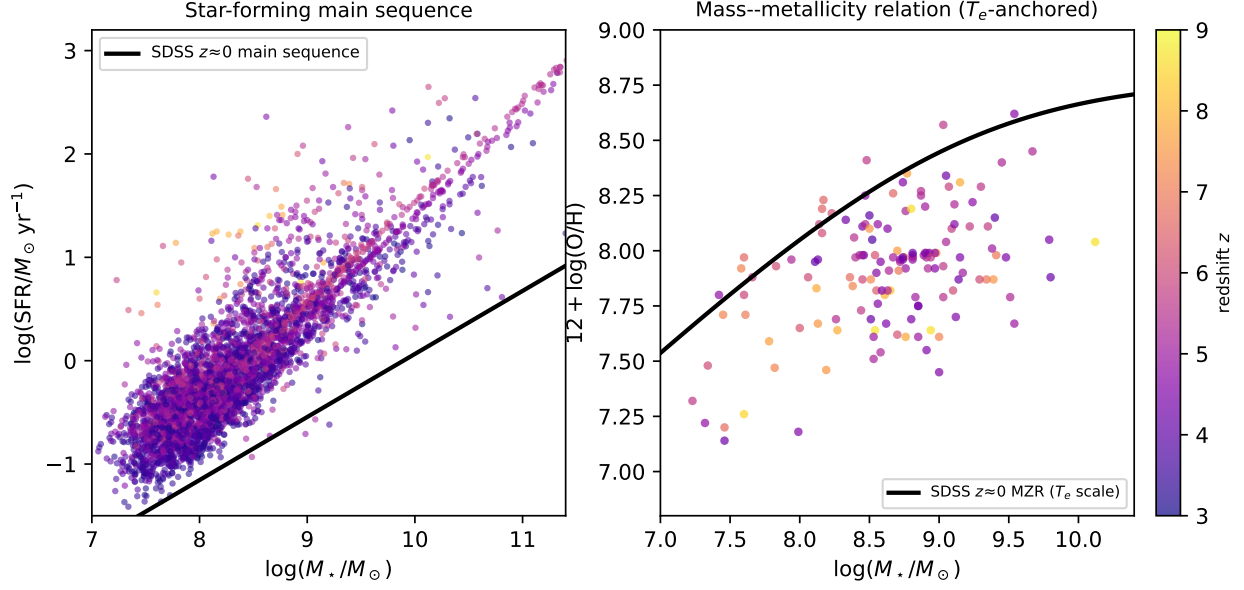


Figure 1. JWST galaxies at $3 < z < 9$ (points, coloured by redshift) overlaid on the SDSS $z \approx 0$ relations (black). *Left:* the star-forming main sequence; high- z galaxies sit systematically above the local sequence. *Right:* the mass-metallicity relation; high- z galaxies lie below the local relation, with large scatter.

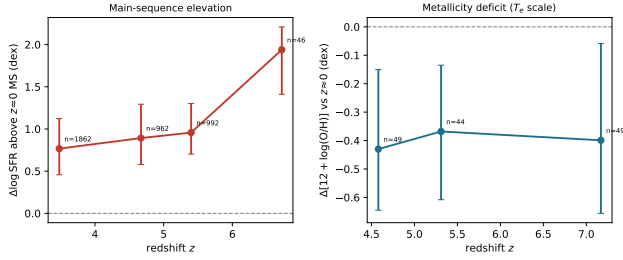


Figure 2. Evolution of the offsets from the $z \approx 0$ relations. *Left:* main-sequence elevation $\Delta \log \text{SFR}$ grows with redshift. *Right:* the metallicity deficit $\Delta_{\text{O}/\text{H}}$ is ≈ -0.4 dex (T_e scale) and nearly flat across $z \approx 4$ –7. Error bars are 16–84th percentiles; n per bin annotated.

REFERENCES

- Brinchmann, J., et al. 2004, MNRAS, 351, 1151
 Curti, M., et al. 2024, A&A, 684, A75
 Kewley, L. J., & Ellison, S. L. 2008, ApJ, 681, 1183
 Lisiecki, K., et al. 2025, A&A, 708, A235
 Moustakas, J., et al. 2011, arXiv:1112.3300
 Nakajima, K., et al. 2023, ApJS, 269, 33
 Noeske, K. G., et al. 2007, ApJ, 660, L43
 Sanders, R. L., et al. 2021, ApJ, 914, 19
 Speagle, J. S., et al. 2014, ApJS, 214, 15
 Tremonti, C. A., et al. 2004, ApJ, 613, 898